INCH-POUND

MIL-I-24768/9
25 February 1992

SUPERSEDING
MIL-P-15047C
11 June 1971

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, NYLON-FABRIC-BASE, PHENOLIC-RESIN (NPG)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a woven nylon cloth substrate properly impregnated and bonded with phenolic-resin matrix and processed to conform to this specification.

Intended use: The laminated material covered by this specification is a moisture and humidity resistant, plasic laminate designed for electronic applications. It has good impact strength but poor heat, flame, and arc resistance. This material shows severe decrease in flexural strength at temperatures above 75 °C.

Forms: Sheets are the only recommended forms of laminated material covered by this specification.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/9-S-1	0.010	0.003
M24768/9-S-2	0.015	0.0035
M24768/9-S-3	0.020	0.004
M24768/9-S-4	0.025	0.045
M24768/9-S-5	0.031	0.0065
M24768/9-S-6	0.047	0.0075
M24768/9-S-7	0.062	0.0075
M24768/9-S-8	0.094	0.009
M24768/9-S-9	0.125	0.010
M24768/9-S-10	0.156	0.011
M24768/9-S-11	0.188	0.0125
M24768/9-S-12	0.219	0.014
M24768/9-S-13	0.250	0.015
M24768/9-S-14	0.312	0.024
M24768/9-S-15	0.375	0.032
M24768/9-S-16	0.438	0.040
M24768/9-S-17	0.500	0.048
M24768/9-S-18	0.625	0.054
M24768/9-S-19	0.750	0.058
M24768/9-S-20	0.875	0.062
M24768/9-S-21	1.000	0.066
M24768/9-S-22	–[1]	–[2]

[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/9-S-22-0.8125.

[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Performance requirements: Performance requirements for laminated sheets shall be as specified in table II.

The following performance requirements do not apply to this plastic laminate:

a. Arc resistance
b. Flammability
c. Compressive strength
d. Silicone content
e. Tracking resistance
f. Dielectric strength
g. Thermal endurance
h. Specific gravity
i. Volume resistivity
j. Surface resistivity.

Testing: The laminated material covered by this specification shall be tested under qualification inspection requirements.

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TABLE II. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[2]	A D-48/50	4.6.4	Min. kV	60.0 40.0	60.0 40.0	60.0 40.0	60.0 40.0	60.0 40.0	60.0 40.0	60.0 40.0	60.0 40.0	50.0 30.0	
Dielectric constant	A D-24/23 D-48/50	4.6.6	Max.	3.90 4.20	3.90 4.00	3.90 4.00	3.90 3.90 4.00	4.20 4.30	4.20 4.30	4.30 4.40	4.40 4.50	4.40 4.50	4.50
Dissipation factor	A D-24/23 D-48/50	4.6.6	Max.	0.038 0.045	0.038 0.041	0.038 0.040	0.038 0.039 0.045	0.038 0.039	0.038 0.039	0.038 0.039	0.038 0.039	0.038 0.039	0.038 0.039
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistivity	C-96/35/90	4.6.8	Min. megohm										
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[2] Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft–lbs. per inch				3.0 2.0	3.0 2.0	3.0 2.0	3.0 2.0	3.0 2.0	3.0 2.0	3.0 2.0

See footnotes at the end of table.

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TABLE II. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Flexural strength	A	4.6.12	Min. psi										
Lengthwise:				10,000	10,000	10,000	10,000	9,500	9,000	9,000	9,000	8,000	8,000
Crosswise:				9,500	9,500	9,500	9,500	9,000	8,500	8,500	8,500	7,500	7,500
Thermal endurance Flexural strength:		4.6.13	Min. psi										
Bonding strength	A	4.6.15	Min. pounds							1,000	1,000	1,000	1,000
	D-48/50									1,000	1,000	1,000	1,000
Water absorption	D_1-24/23	4.6.17	Max. percent	0.90	0.60	0.50	0.40	0.40	0.38	0.35			
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Maximum thickness tested shall be 2.000 inches.

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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
(Project 5970-1015-09)

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